

Министерство образования Республики Беларусь
Учреждение образования
«Брестский государственный технический университет»
Кафедра ИИТ

Лабораторная работа №3

По дисциплине «Обработка изображений в интеллектуальных системах»

Тема: «Обучение детекторов объектов»

Выполнил:

Студент 4 курса

Группы ИИ-26

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Проверила:

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Брест 2026

Цель: осуществлять обучение нейросетевого детектора для решения задачи обнаружения заданных объектов

Ход работы

1. Базируясь на своем варианте, ознакомится с выборкой для обучения детектора, выполнить необходимые преобразования данных для организации процесса обучения (если это нужно!);
2. Для заданной архитектуры нейросетевого детектора организовать процесс обучения для своей выборки. Оценить эффективность обучения на тестовой выборке (mAP);
3. Реализовать визуализацию работы детектора из пункта 1 (обнаружение знаков на отдельных фотографиях из сети Интернет);
4. Оформить отчет по выполненной работе, залить исходный код и отчет в соответствующий репозиторий на github.

В-т	Детектор	Датасет
7	YOLO12n	Люди: https://universe.roboflow.com/leo-ueno/people-detection-o4rdr/dataset/10

Код программы:

Результат программы:

Классы из экспорта: ['person']

train: 15210 фото, 100665 рамок, 1922 фоновых фото; преобразовано полигонов: 34

val: 1431 фото, 10683 рамок, 61 фоновых фото; преобразовано полигонов: 23

test: 760 фото, 2783 рамок, 22 фоновых фото; преобразовано полигонов: 0

PyTorch: 2.11.0+cu128; CUDA доступна: True; устройство: 0

Обучение YOLO12n: device=0, epochs=8, batch=4

Ultralytics 8.4.174 Python-3.13.3 torch-2.11.0+cu128 CUDA:0 (NVIDIA GeForce RTX 3050 Laptop GPU, 4096MiB)

engine\trainer: agnostic_nms=False, amp=False, angle=1.0, augment=False, auto_augment=randaugument, batch=4, bgr=0.0, box=7.5, cache=False, cfg=None, channels_last=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0, cls_remap=True, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\data.yaml, degrees=0.0, deterministic=True, device=0, dfl=1.5, dgrad=0.5, dis=6.0, distill_model=None, dlam=1.0, dlog=1.0, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=8, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, iou=0.7, kobj=1.0, line_width=None, lr0=0.001, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo12n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=train-2, nbs=64, nms=None, opset=None, optimize=False, optimizer=AdamW, overlap_mask=True, patience=15, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs, quantize=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\train-2, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=42, shear=0.0, show=False,

show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=None, tracker=tracktrack.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, workers=0, workspace=None
Overriding model.yaml nc=80 with nc=1

	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	6640	ultralytics.nn.modules.block.C3k2	[32, 64, 1, False, 0.25]
3	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
4	-1	1	26080	ultralytics.nn.modules.block.C3k2	[64, 128, 1, False, 0.25]
5	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
6	-1	2	180864	ultralytics.nn.modules.block.A2C2f	[128, 128, 2, True, 4]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8	-1	2	689408	ultralytics.nn.modules.block.A2C2f	[256, 256, 2, True, 1]
9	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
10	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
11	-1	1	86912	ultralytics.nn.modules.block.A2C2f	[384, 128, 1, False, -1]
12	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
13	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
14	-1	1	24000	ultralytics.nn.modules.block.A2C2f	[256, 64, 1, False, -1]
15	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
16	[-1, 11]	1	0	ultralytics.nn.modules.conv.Concat	[1]
17	-1	1	74624	ultralytics.nn.modules.block.A2C2f	[192, 128, 1, False, -1]
18	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
19	[-1, 8]	1	0	ultralytics.nn.modules.conv.Concat	[1]
20	-1	1	378880	ultralytics.nn.modules.block.C3k2	[384, 256, 1, True]
21	[14, 17, 20]	1	430867	ultralytics.nn.modules.head.Detect	[1, 16, None, [64, 128, 256]]

YOLOv12n summary: 272 layers, 2,568,243 parameters, 2,568,227 gradients, 7.5 GFLOPs

Remapped 1/1 cls head rows from pretrained weights by class name

Transferred 643/691 items from pretrained weights

Freezing layer 'model.21.dfl.conv.weight'

WARNING train: Slow image access detected (ping: 0.10.0 ms, read: 26.526.5 MB/s, size: 113.2 KB).

Use local storage instead of remote/mounted storage for better performance. See

<https://docs.ultralytics.com/guides/model-training-tips>

train: Scanning C:\Users\fhdsjfbjdjsg\Desktop\PythonProject33\.venv\prepared_dataset\train\labels...

15210 images, 1922 backgrounds, 0 corrupt: 100% ————— 15210/15210 297.3it/s
51.2s

train:

C:\Users\fhdsjfbjdjsg\Desktop\PythonProject33\.venv\prepared_dataset\train\images\2008_003311.jpg.r
f.01d53f54203b927cbb892628f6447d8e.jpg: 7 duplicate labels removed

train:

C:\Users\fhdsjfbjdjsg\Desktop\PythonProject33\.venv\prepared_dataset\train\images\2008_003493.jpg.r
f.631ad603a7fb17307ef23d4d541abd7c.jpg: 1 duplicate labels removed

train:

C:\Users\fhdsjfbjdjsg\Desktop\PythonProject33\.venv\prepared_dataset\train\images\2008_005646.jpg.r
f.e8a515492d9ca7ac40f764eb5351265c.jpg: 1 duplicate labels removed

train:
C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\train\images\2008_006285_jpg.rf.bda2de2bed84a627472b264630e48d6b.jpg: 7 duplicate labels removed

train:
C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\train\images\2008_006924_jpg.rf.3f6ad257e43f80afdc40cd22464f1212.jpg: 1 duplicate labels removed

train:
C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\train\images\2010_002985_jpg.rf.69395c8b29f9eb0378f29187ae32adff.jpg: 1 duplicate labels removed

train:
C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\train\images\2010_003239_jpg.rf.ac5bc7fcc17a9f44480351566f9ad48b.jpg: 7 duplicate labels removed

train: New cache created:
C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\train\labels.cache
WARNING val: Slow image access detected (ping: 0.10.0 ms, read: 16.917.3 MB/s, size: 71.1 KB). Use local storage instead of remote/mounted storage for better performance. See <https://docs.ultralytics.com/guides/model-training-tips>
val: Scanning C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\val\labels... 1431 images, 61 backgrounds, 0 corrupt: 100% 1431/1431 289.5it/s 4.9s

val:
C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\val\images\GX010023_frame_00025_right_jpg.rf.3255febc597d78d5d6e4bde455ba7b2a.jpg: 23 duplicate labels removed

val: New cache created:
C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\val\labels.cache
optimizer: AdamW(lr=0.001, momentum=0.937) with parameter groups 113 weight(decay=0.0), 120 weight(decay=0.0005), 119 bias(decay=0.0)
Plotting labels to C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\train-2\labels.jpg...
Using 15210 train, 1431 val images for fraction=1.0 at imgsz=640
Using 0 dataloader workers
Logging results to C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\train-2
Starting training for 8 epochs...
Closing dataloader mosaic

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/8	1.47G	1.73	1.833	1.567	18	640: 100%
3803/3803	6.9it/s	9:14				
Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
all	1431	10660	0.615	0.396	0.423	0.19

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
2/8	1.7G	1.654	1.692	1.526	0	640: 100%
3803/3803	7.0it/s	9:02				
Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
all	1431	10660	0.608	0.407	0.422	0.178

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
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3/8	1.7G	1.58	1.566	1.48	14	640: 100%		
3803/3803 6.8it/s 9:20								
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	
	179/179	9.9it/s	18.1s					
	all	1431	10660	0.662	0.446	0.485	0.234	

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances	Size		
4/8	1.7G	1.518	1.48	1.429	2	640: 100%		
3803/3803 6.9it/s 9:08								
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	
	179/179	9.8it/s	18.3s					
	all	1431	10660	0.668	0.475	0.517	0.26	

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances	Size		
5/8	1.7G	1.453	1.376	1.386	4	640: 100%		
3803/3803 7.1it/s 8:59								
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	
	179/179	9.8it/s	18.2s					
	all	1431	10660	0.642	0.483	0.519	0.246	

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances	Size		
6/8	1.7G	1.395	1.296	1.344	7	640: 100%		
3803/3803 7.1it/s 8:58								
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	
	179/179	9.8it/s	18.3s					
	all	1431	10660	0.692	0.486	0.532	0.266	

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances	Size		
7/8	1.7G	1.351	1.223	1.317	79	640: 100%		
3803/3803 7.0it/s 9:01								
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	
	179/179	9.7it/s	18.4s					
	all	1431	10660	0.726	0.484	0.509	0.281	

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances	Size		
8/8	1.7G	1.295	1.157	1.286	4	640: 100%		
3803/3803 7.1it/s 8:59								
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	
	179/179	9.7it/s	18.4s					
	all	1431	10660	0.745	0.478	0.528	0.294	

8 epochs completed in 1.253 hours.

Optimizer stripped from C:\Users\fhdsjfbdsjg\Desktop\PythonProject33\venv\runs\train-2\weights\last.pt, 5.5MB

Optimizer stripped from C:\Users\fhdsjfbdsjg\Desktop\PythonProject33\venv\runs\train-2\weights\best.pt, 5.5MB

Validating C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\train-2\weights\best.pt...
 Ultralytics 8.4.174 Python-3.13.3 torch-2.11.0+cu128 CUDA:0 (NVIDIA GeForce RTX 3050 Laptop GPU, 4096MiB)
 YOLOv12n summary (fused): 159 layers, 2,556,923 parameters, 0 gradients, 7.3 GFLOPs

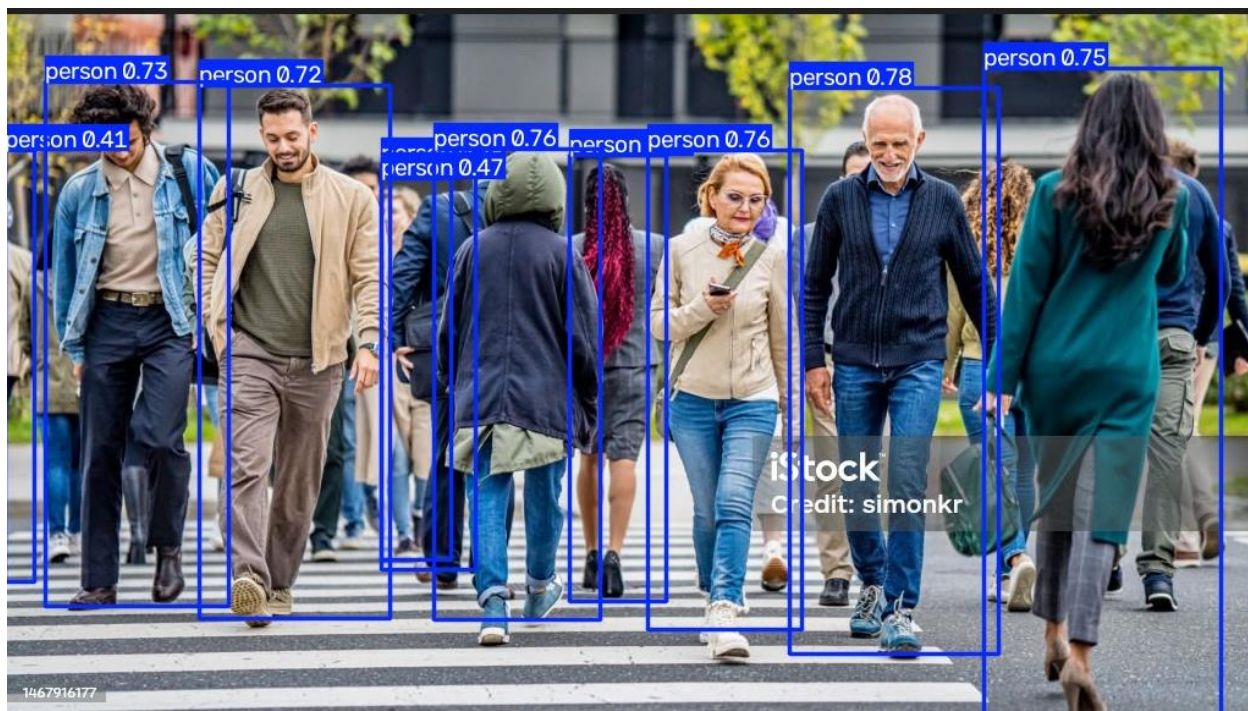
Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
all	1431	10660	0.745	0.478	0.528	0.294

Speed: 0.2ms preprocess, 5.1ms inference, 0.0ms loss, 0.9ms postprocess per image
 Results saved to C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\train-2
 Лучшая модель: C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\train-2\weights\best.pt
 Ultralytics 8.4.174 Python-3.13.3 torch-2.11.0+cu128 CUDA:0 (NVIDIA GeForce RTX 3050 Laptop GPU, 4096MiB)
 YOLOv12n summary (fused): 159 layers, 2,556,923 parameters, 0 gradients, 7.3 GFLOPs
 WARNING val: Slow image access detected (ping: 0.10.0 ms, read: 8.82.3 MB/s, size: 38.0 KB). Use local storage instead of remote/mounted storage for better performance. See <https://docs.ultralytics.com/guides/model-training-tips>
 val: Scanning C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\test\labels... 760 images, 22 backgrounds, 0 corrupt: 100% 760/760 307.2it/s 2.5s
 val: New cache created:
 C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\prepared_dataset\test\labels.cache

Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
all	760	2783	0.795	0.745	0.791	0.488

Speed: 0.3ms preprocess, 6.3ms inference, 0.0ms loss, 0.9ms postprocess per image
 Results saved to C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\test

РЕЗУЛЬТАТЫ НА TEST:
 precision: 0.7955 (79.55%)
 recall: 0.7452 (74.52%)
 mAP50: 0.7914 (79.14%)
 mAP50-95: 0.4877 (48.77%)
 0: 384x640 10 persons, 49.7ms
 Speed: 3.0ms preprocess, 49.7ms inference, 4.8ms postprocess per image at shape (1, 3, 384, 640)
 Results saved to C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\predict
 1 label saved to C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\predict\labels
 istockphoto-1467916177-1024x1024.jpg: найдено людей — 10; результат:
 C:\Users\fhdgsjfbdjsg\Desktop\PythonProject33\venv\runs\predict



Labels:

0 0.711127 0.507298 0.167393 0.804411 0.775482
 0 0.57548 0.534395 0.122804 0.680027 0.757923
 0 0.409731 0.526151 0.13311 0.66812 0.756348
 0 0.877651 0.537402 0.189419 0.919098 0.747065
 0 0.105962 0.468063 0.145665 0.744988 0.733455
 0 0.231795 0.479732 0.151884 0.756878 0.717404
 0 0.490329 0.517366 0.0778061 0.633123 0.615489
 0 0.329224 0.503751 0.0552285 0.541423 0.469547
 0 0.0121014 0.499214 0.0239433 0.612689 0.410453
 0 0.339155 0.500674 0.0750195 0.576394 0.27954

Вывод: осуществил обучение нейросетевого детектора для решения задачи обнаружения заданных объектов.